

# AHSANUL AMIN

## Electrical & Electronic Engineering Graduate

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### PROFESSIONAL SUMMARY

Electrical & Electronic Engineering graduate from CUET with academic and internship experience in power systems, electrical distribution, power electronics, and engineering analysis. Gained practical exposure to utility operations through DESCO internship and developed skills in MATLAB/Simulink, PowerWorld Simulator, AutoCAD, and technical documentation. Interested in high-voltage substation design, electrical systems, and power engineering projects with a strong willingness to learn and contribute.

### EDUCATION

#### B.Sc. in Electrical & Electronic Engineering 2022 – 2026

Chittagong University of Engineering & Technology (CUET), Chattogram

Relevant coursework: Power Systems, Switchgear and Protection, Electrical Machines, Power Electronics, Electric Circuits, Control Systems, Renewable Energy Systems, Digital Electronics.

#### Higher Secondary Certificate (H.S.C.) – Science 2020

Government Science College, Dhaka | GPA: 5.00/5.00

#### Secondary School Certificate (S.S.C.) – Science 2018

Khilgaon Government High School, Dhaka | GPA: 5.00/5.00

### PROJECT EXPERIENCE

#### Hybrid SEPIC-Zeta Bidirectional Converter with Adaptive Fuzzy Current Control for G2V/V2G Applications (Undergraduate Thesis)

- Designed and analyzed a bidirectional power converter system involving power electronics, control systems, simulation, and grid integration concepts.
- Prepared technical documentation and performed engineering analysis throughout the project.

#### Solar PV System Design & Simulation on PVsyst

- Learned solar PV system design concepts and performed PV system analysis using PVsyst.
- Studied system sizing, energy yield estimation, and performance evaluation.

#### An Annular Photonic Crystal Fiber SPR Sensor for Density-Based Refractive-Index Detection of Olive and Soybean Oil (Biomedical Research Project Using COMSOL Multiphysics)

- Conducted simulation-based analysis using COMSOL Multiphysics and developed research and documentation skills.

#### Touch-Based LED Lighting Control System (Course Project)

- Developed a practical electronics project involving circuit design and implementation.

#### Line Follower Robot (Team Project)

- Built and tested a robotics project through teamwork and hands-on implementation.

#### VLSI Design Project – 4-bit Up/Down Counter | Cadence Virtuoso

- Designed and implemented a 4-bit up/down counter using J-K flip-flops following a full-custom VLSI design flow.

## INTERNSHIP EXPERIENCE

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### Dhaka Electric Supply Company (DESCO)

2025 / 2 Weeks

DESCO, Dhaka, Bangladesh

- Gained practical exposure to electricity distribution systems, utility operations, and field-level engineering activities.
- Observed electrical equipment, protection systems, and operational procedures in power distribution networks.
- Developed understanding of industrial electrical practices through technical observation and field exposure.

## TOOLS & SOFTWARE

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### Microsoft Office

MS Excel, MS Word, MS PowerPoint

### Engineering Tools

AutoCAD, MATLAB/Simulink, Cadence Virtuoso, PowerWorld Simulator, PVsyst, COMSOL Multiphysics, Proteus

## LEADERSHIP & CAMPUS ENGAGEMENT

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- Served as Class Representative (CR), coordinating academic communication between students and faculty.
- Executive Member & Treasurer, IEEE Student Branch CUET
- Assisted in event launches and volunteer activities, supporting coordination, teamwork, and on-ground execution.

## LANGUAGES

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**Bengali** – Native | **English** – Professional Working Proficiency